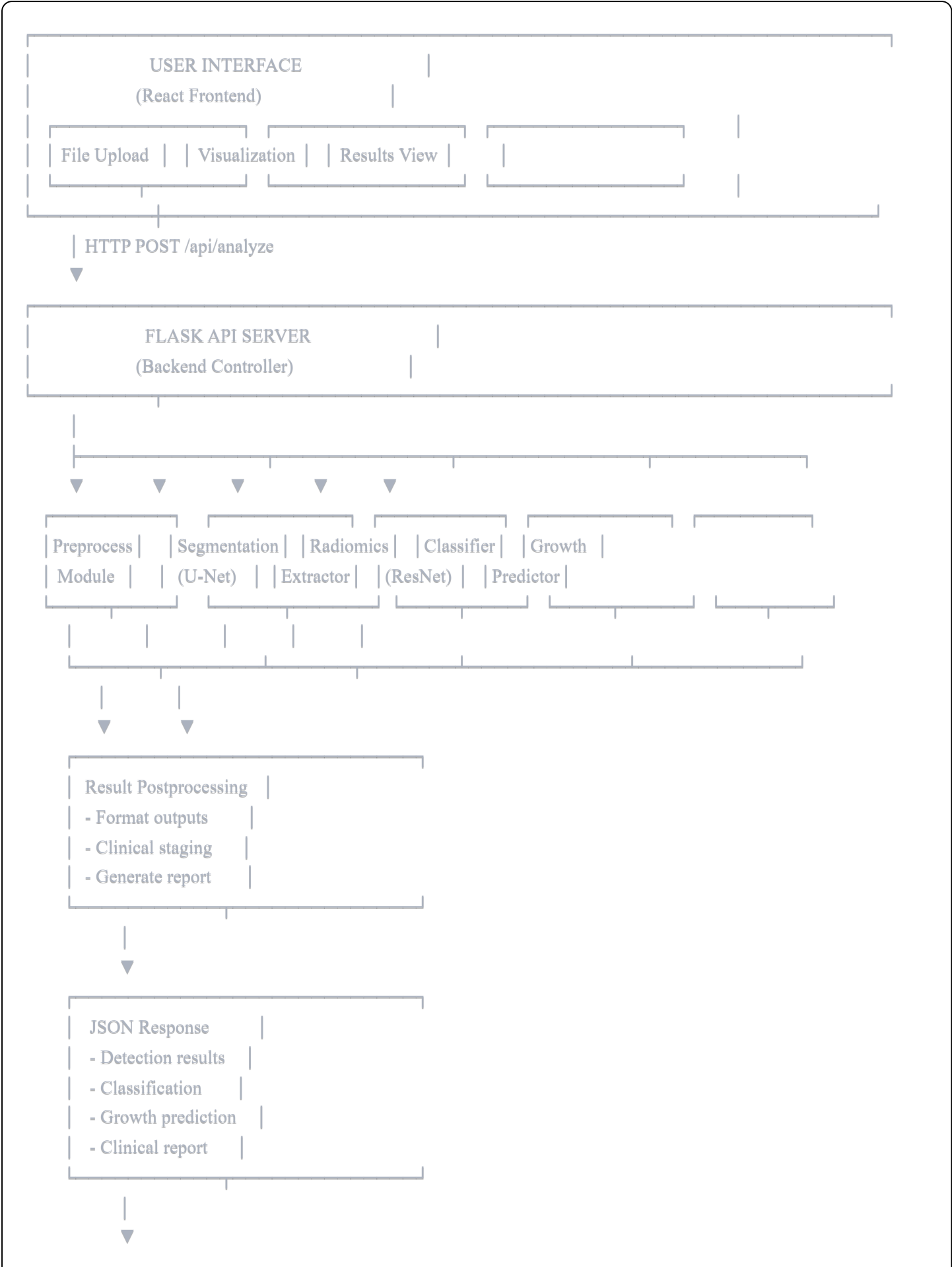


Kidney Tumor Analysis System - Flowchart Diagrams

1. Overall System Architecture Flowchart



- Frontend Display
- 5 Analysis Tabs
- Visualizations
- Downloadable Report

2. Complete Analysis Pipeline Flowchart

START



User Uploads CT Scan
(PNG/JPG/DICOM)



Image Validation

- Check format
- Check size
- Check quality

Invalid



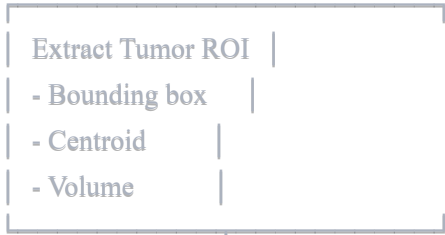
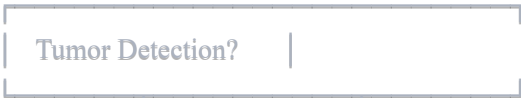
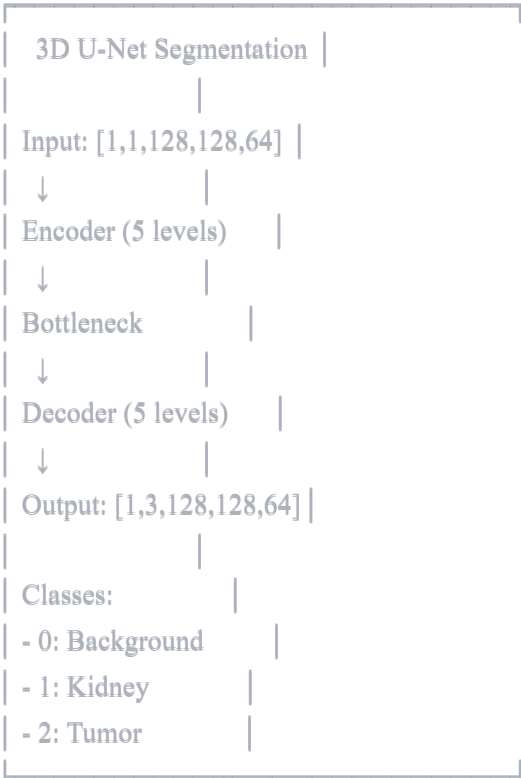
Error Message
Return 400

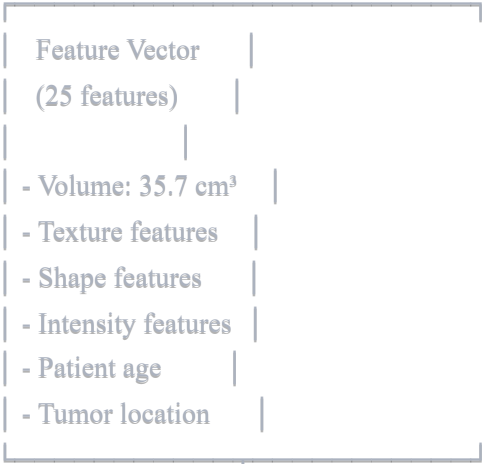
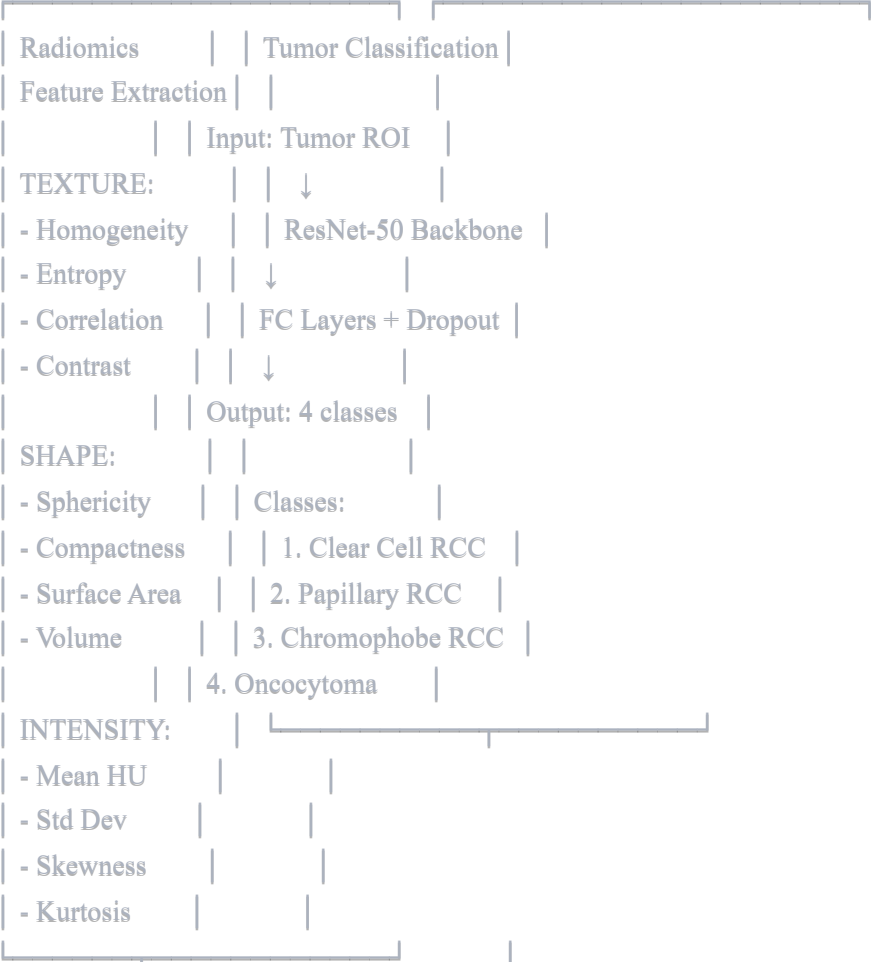
Valid

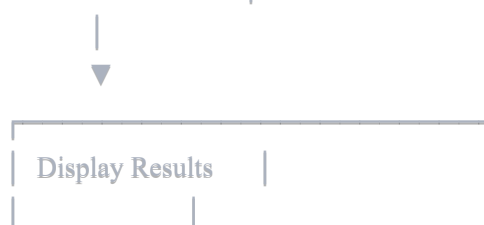
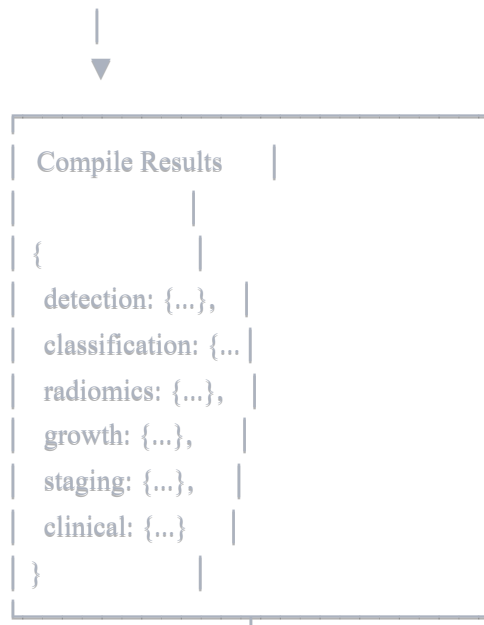
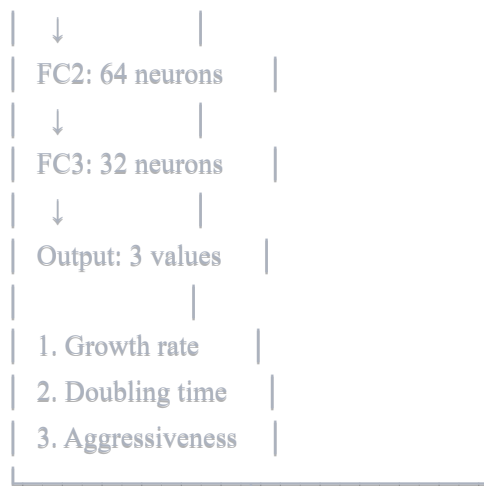


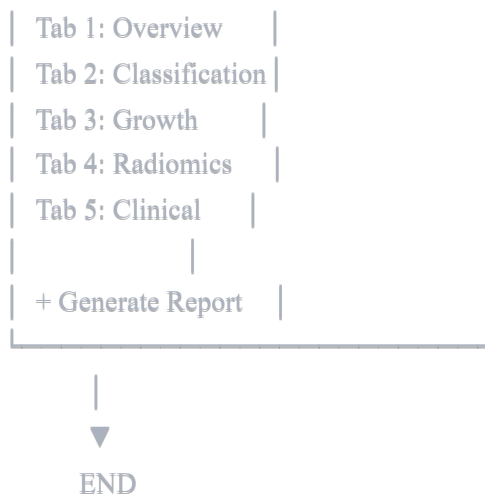
Preprocessing

1. Convert to grayscale
2. Apply HU windowing
3. Normalize intensity
4. Resize (128×128×64)
5. Denoise (optional)
6. Contrast enhancement



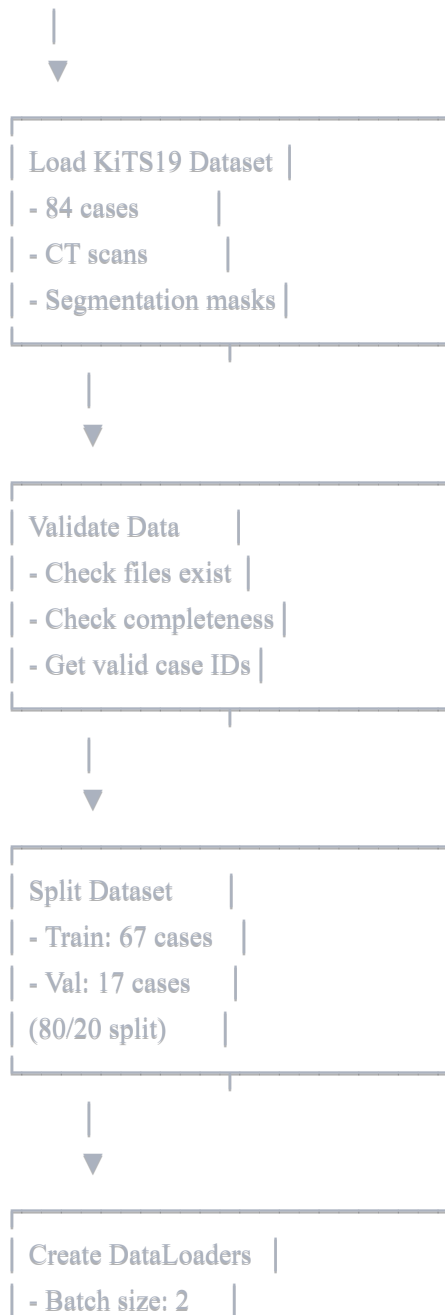






3. Training Pipeline Flowchart

START TRAINING



- Shuffle train

- Num workers: 0



Initialize Model

- 3D U-Net

- 32 init features

- 31M parameters



Setup Training

- Adam optimizer

- LR: 1e-4

- Combined Loss

- LR Scheduler



For epoch

1 to 50



TRAINING PHASE



For each batch

in train_loader



1. Load batch

- Images: 2

- Labels: 2



2. Forward Pass

- `model(images)`
- Get outputs



3. Calculate Loss

- Dice Loss
- CE Loss
- Combined



4. Backward Pass

- `loss.backward()`
- `optimizer.step`



5. Calculate Metrics

- Dice per class
- Running loss



VALIDATION PHASE

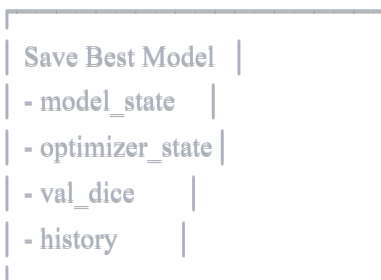
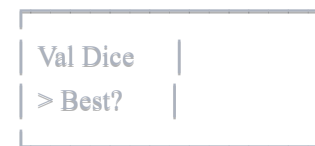
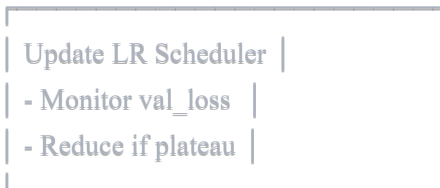
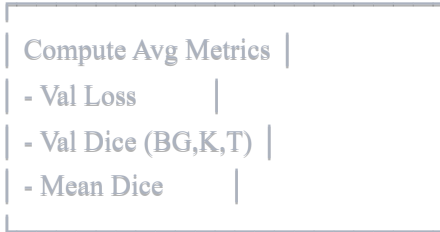


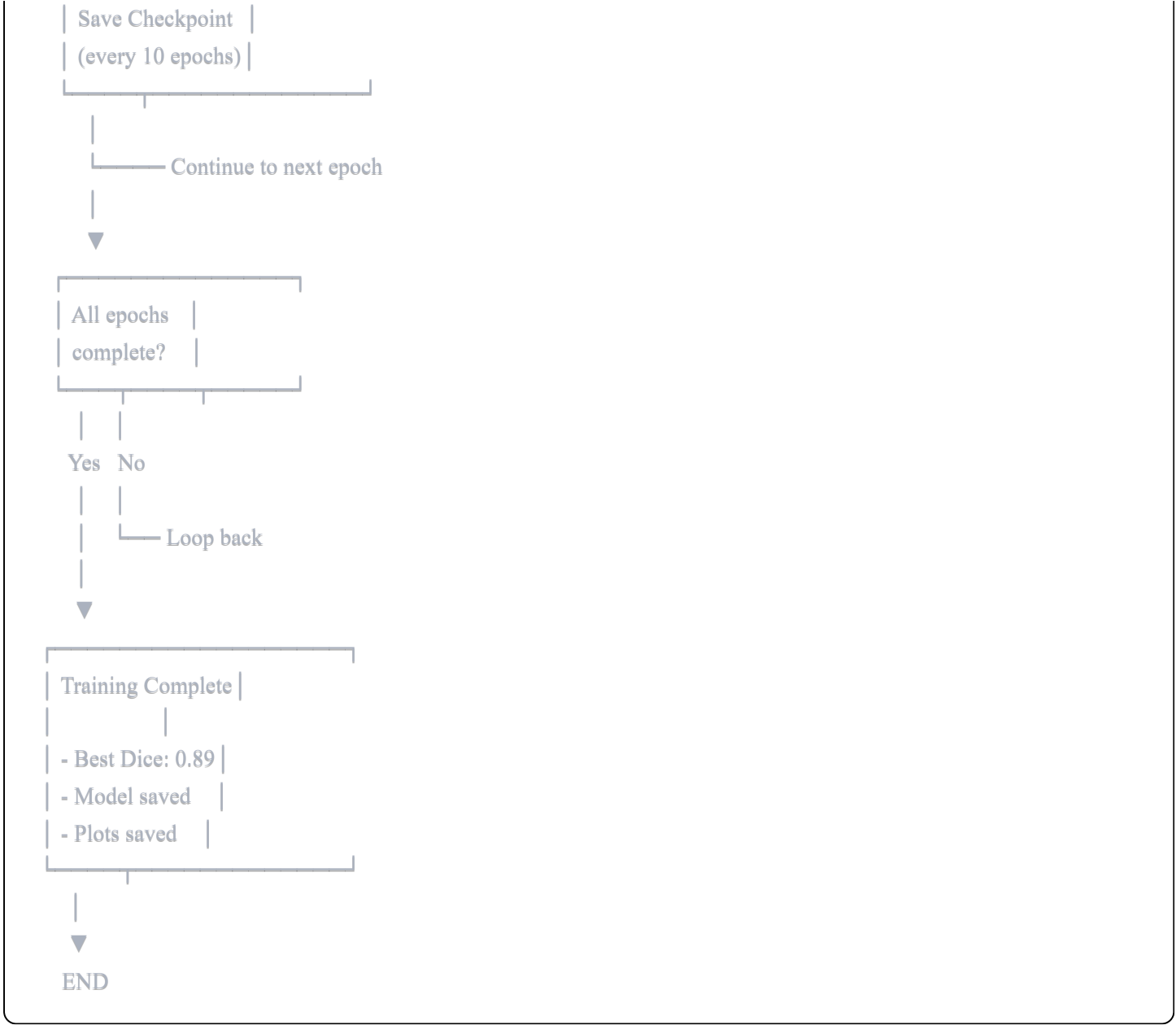
`model.eval()`
`with torch.no_grad()`



For each batch
in `val_loader`

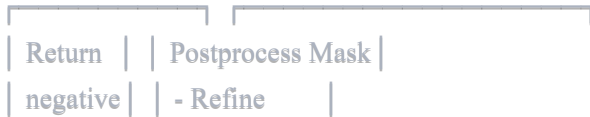
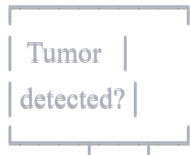
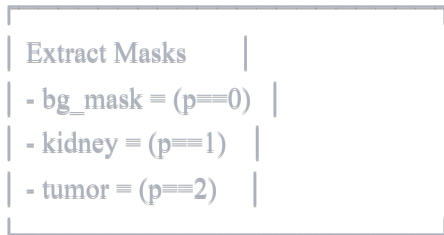
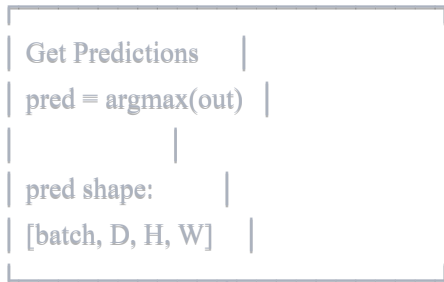
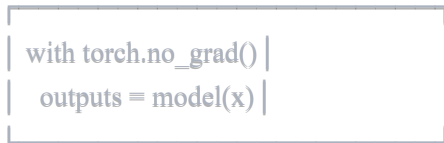
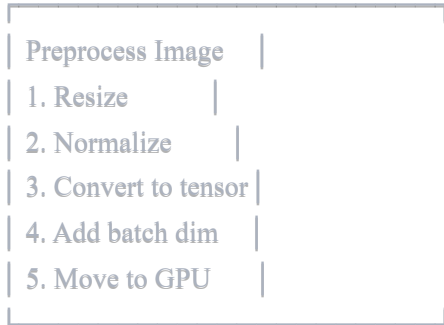


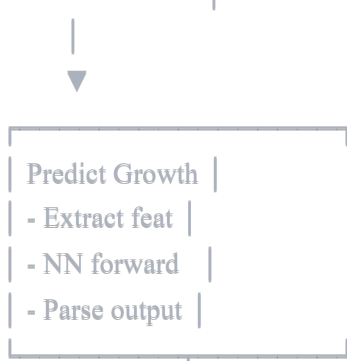
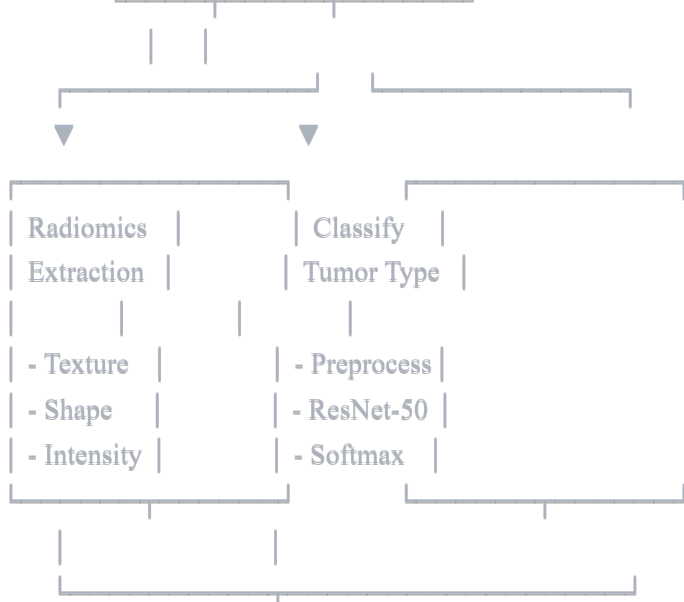
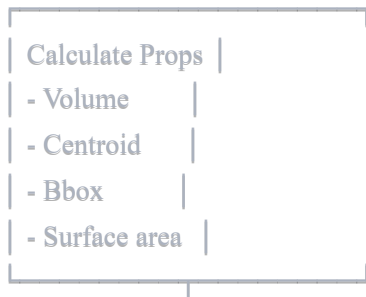
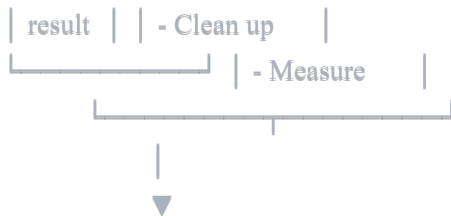


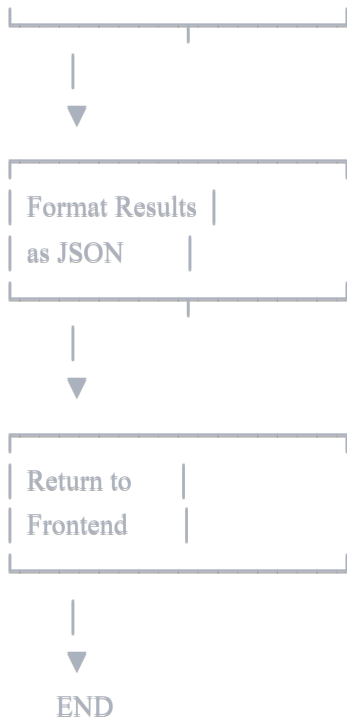


4. Model Inference Flowchart (Real-time Analysis)

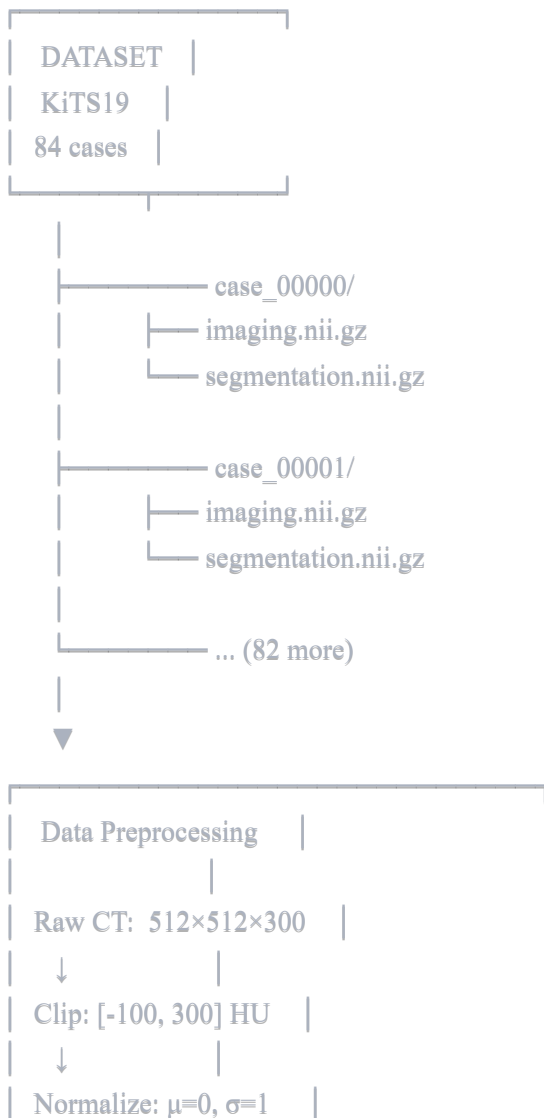


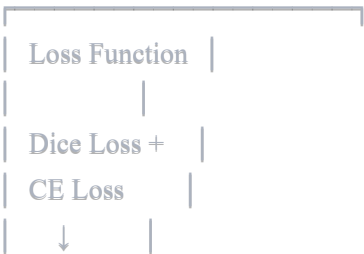
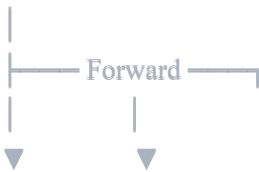
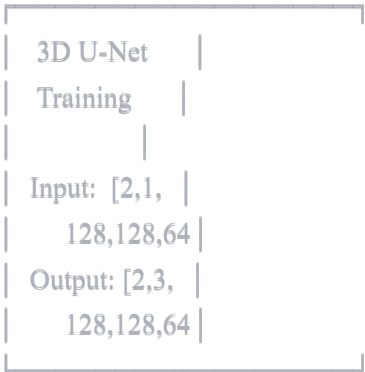
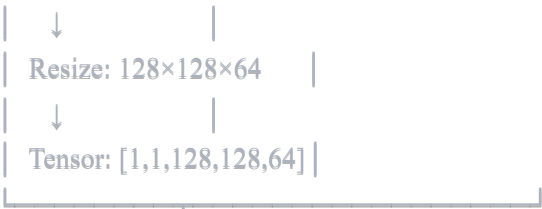






5. Data Flow Diagram







6. Frontend User Flow Diagram

USER OPENS APP



File	
Picker	
Dialog	

| Select File



Preview	
Image	
[Analyze]	

| Click Analyze



Loading	
Spinner	
"Analyzing"	

| (API Call)
| POST /api/analyze



Results	
Received	



Results Dashboard				
Tab1	Tab2	Tab3	Tab4	T5
Active Tab View				
- Charts				
- Metrics				
- Visualizations				
[Download Report]				

Tab 1: Overview

- Detection status
- Tumor size
- Location
- Staging

Tab 2: Classification

- Tumor type
- Confidence
- Subtypes
- Enhancement

Tab 3: Growth

- Growth rate
- Predictions
- Timeline chart
- Aggressiveness

Tab 4: Radiomics

- Texture features
- Shape metrics
- Intensity stats
- Feature plots

Tab 5: Clinical

- Recommendations
- Prognosis
- Treatment options
- Risk assessment



Download	
Report	
(PDF/TXT)	

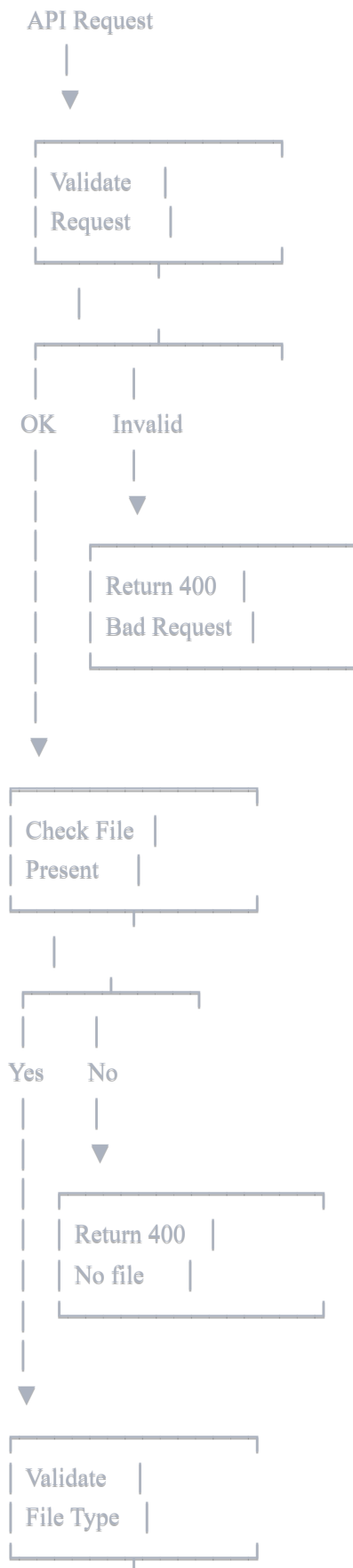


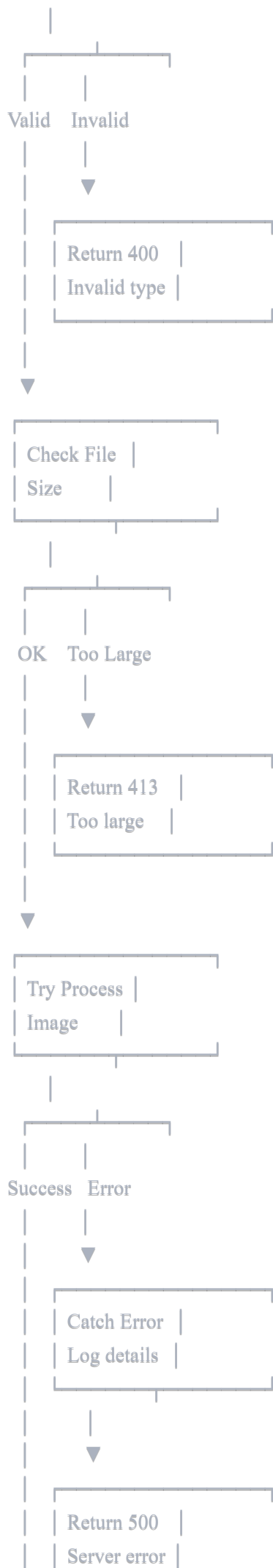
New	
Analysis?	

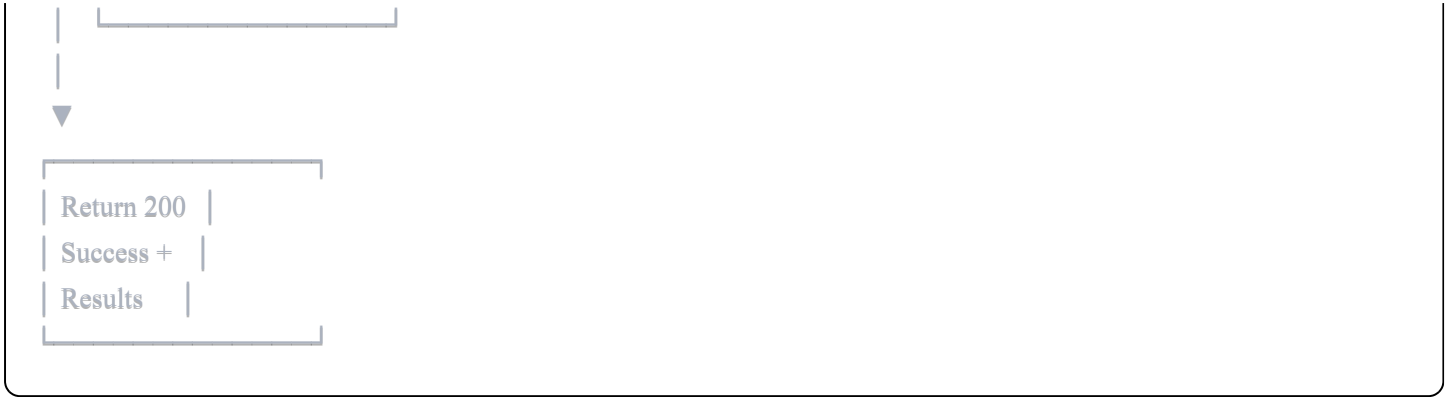
Yes —► Back to Upload



7. Error Handling Flowchart

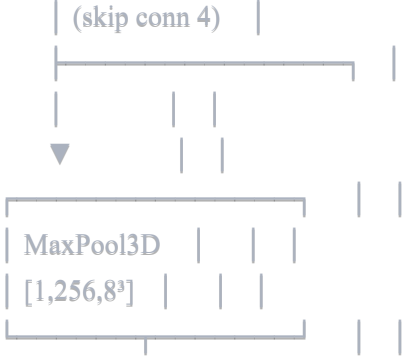
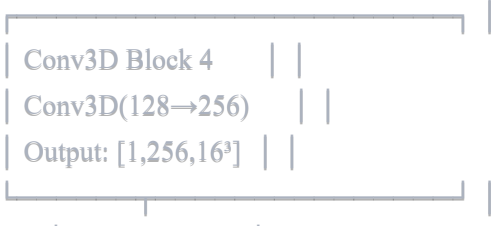
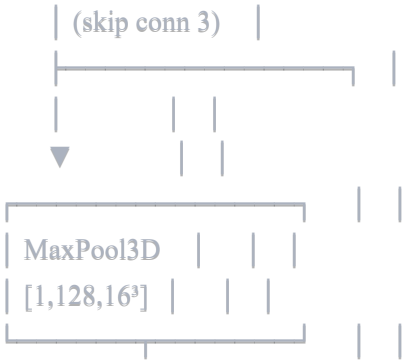
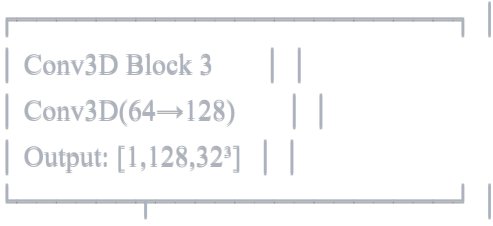
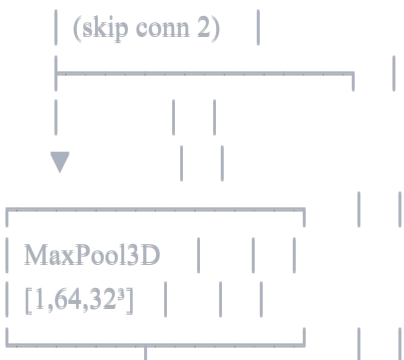
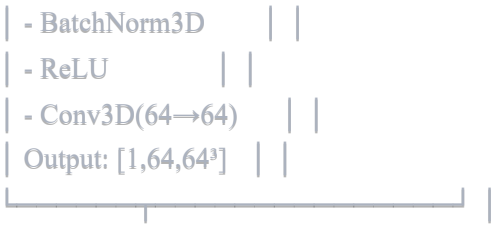




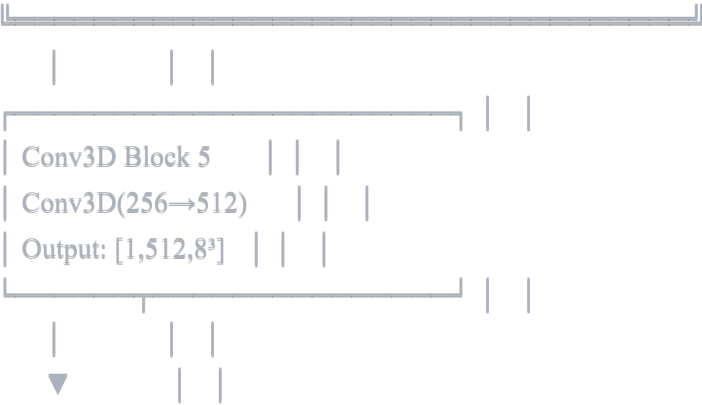


8. Model Architecture Flowchart (3D U-Net Detailed)

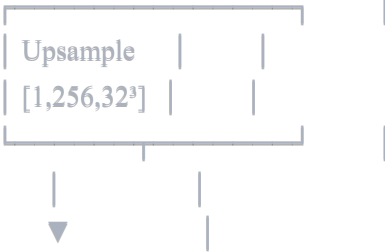
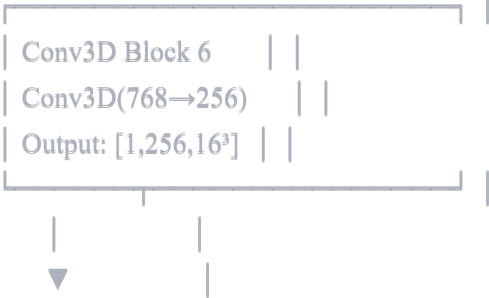
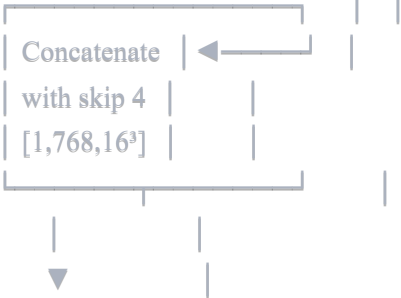
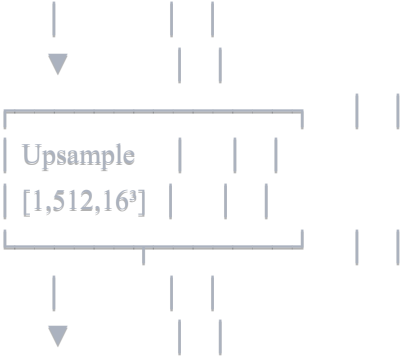


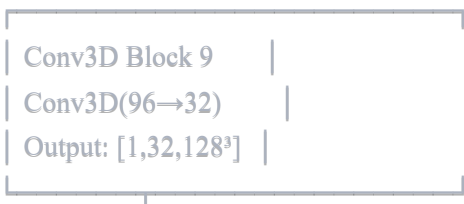
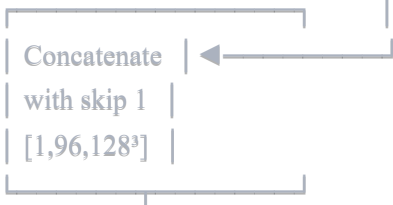
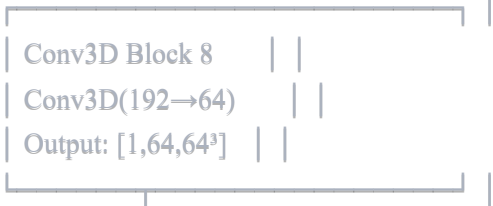
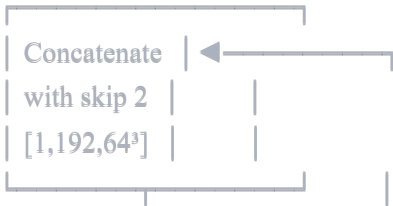
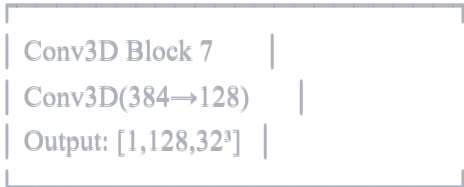


BOTTLENECK



DECODER PATH



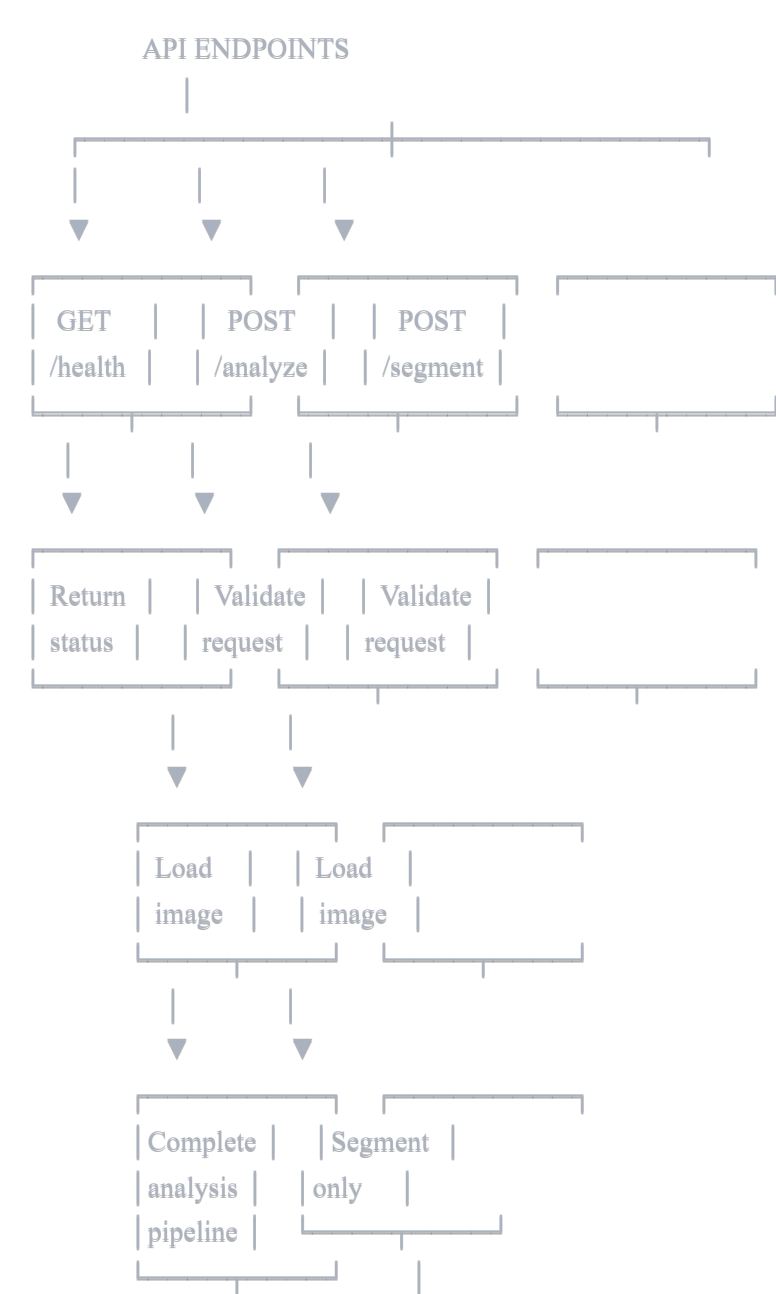


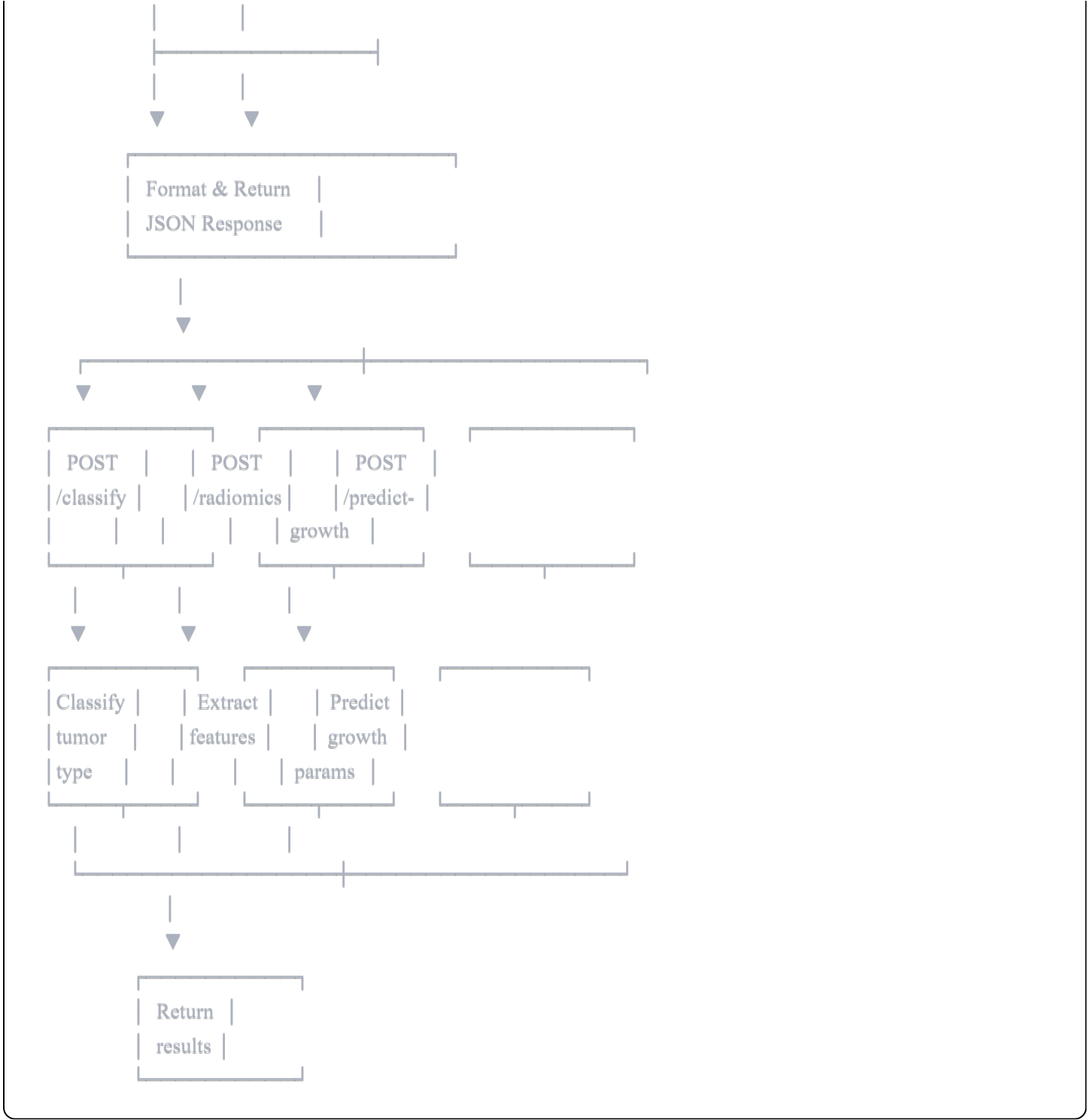


OUTPUT SEGMENTATION

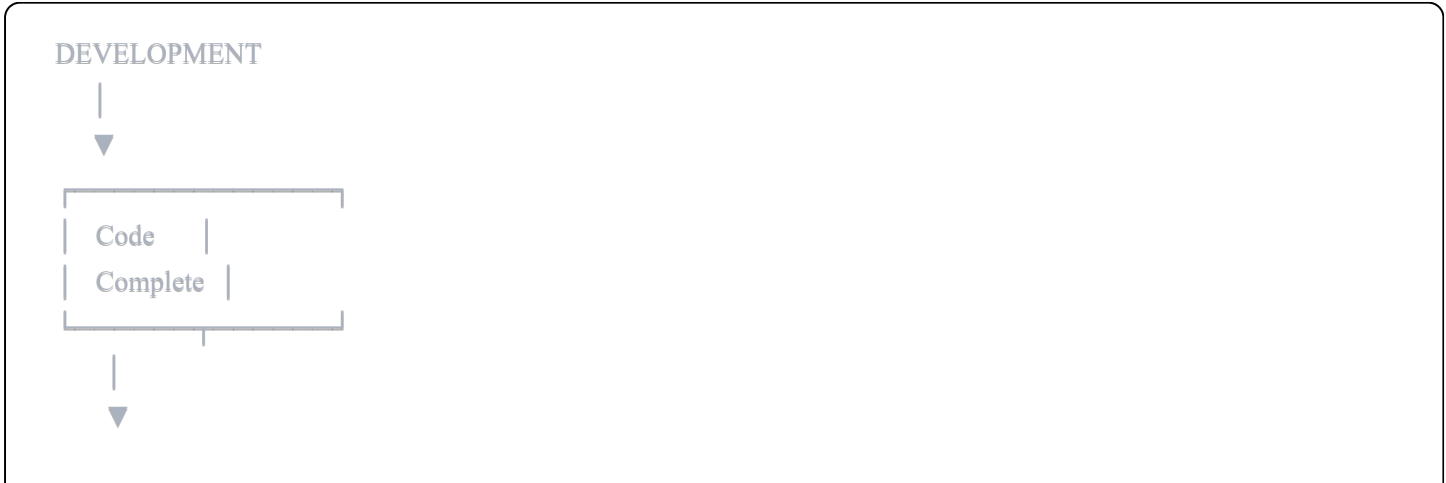
- [1, 3, 128, 128, 64]
- Channel 0: Background
 - Channel 1: Kidney
 - Channel 2: Tumor

9. API Endpoint Flowchart





10. Deployment Workflow



Train
Models



Test
Locally



Pass

Fail



Fix bugs



Back to test



Build
Docker
Images



Backend Image

- Python 3.9
- PyTorch
- Flask
- Models



Frontend Image

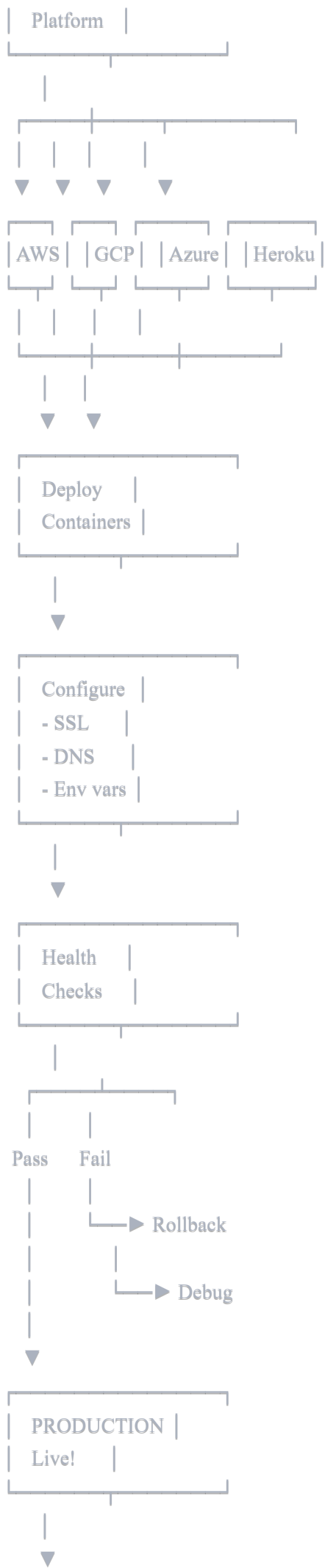
- Node 18
- React build
- Nginx



Push to
Registry
(DockerHub)

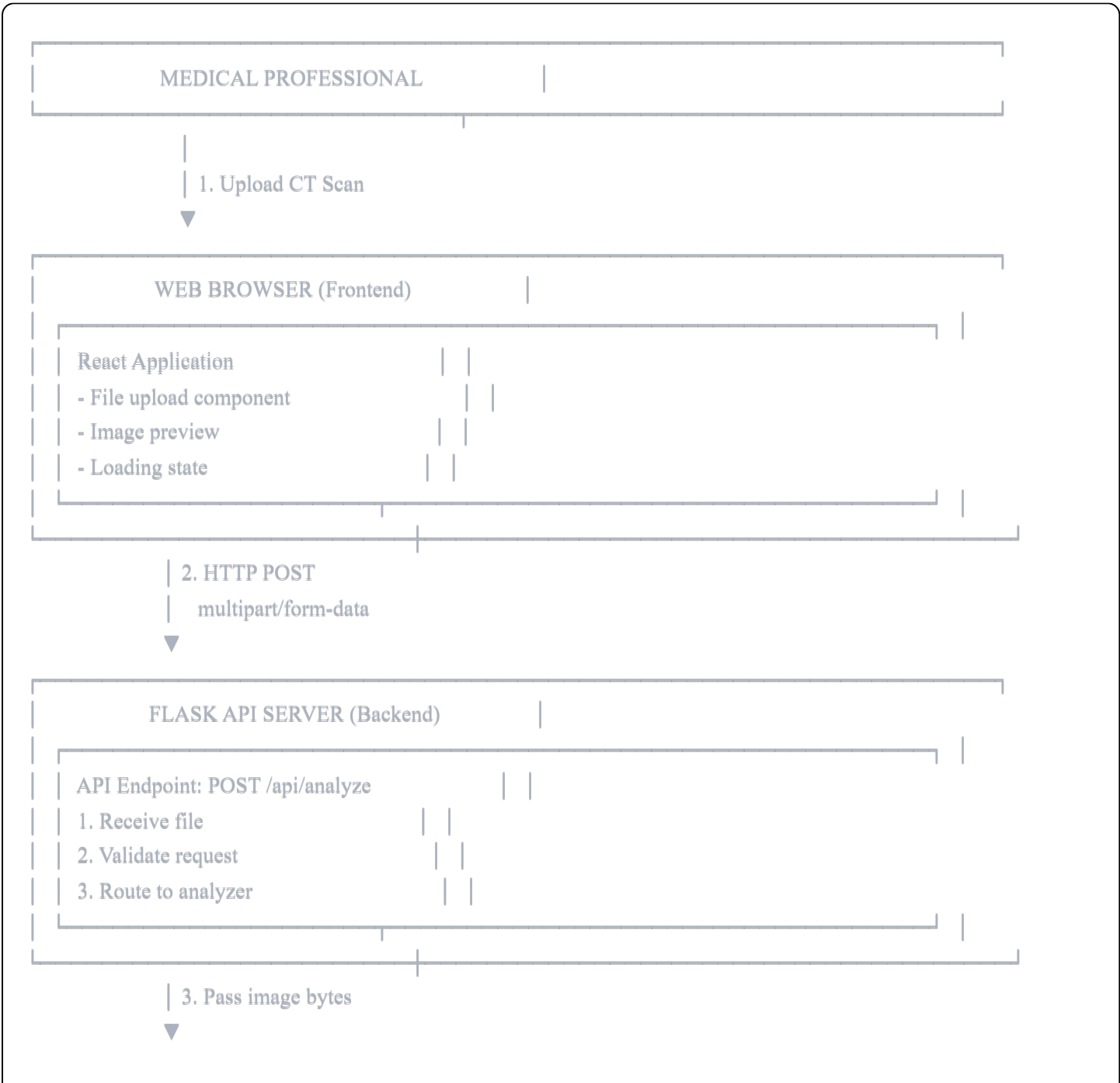


Choose





11. Complete System Data Flow



KIDNEY TUMOR ANALYZER (Core Logic)

Step 1: Preprocessing

- Convert to grayscale
- Apply HU windowing
- Normalize
- Resize to 128³

4. Preprocessed tensor

Step 2: 3D U-Net Segmentation

- Forward pass through model
- Get 3-class predictions
- Extract tumor mask

5. Segmentation mask

Step 3: Post-processing

- Clean up mask
- Remove noise
- Calculate properties

6. Refined mask + metrics

Step 4: Parallel Analysis

Radiomics

- Texture
- Shape
- Intensity

Classification

- ResNet-50 forward
- Get tumor type
- Confidence scores

7. Features + classification

Step 5: Growth Prediction

- Extract 25 features
- Neural network forward
- Predict: rate, time, aggression

8. Growth predictions

Step 6: Clinical Analysis

- TNM staging
- Risk assessment

- Generate recommendations

- Calculate prognosis

9. Complete analysis results

RESULT FORMATTER

Compile JSON Response:

```
{
  detection: {...},
  classification: {...},
  radiomics: {...},
  growth: {...},
  staging: {...},
  clinical: {...}
}
```

10. HTTP 200 OK

JSON response

FRONTEND (React)

Parse Results & Display:

Tab 1: Overview

- Tumor detected: Yes

- Size: 35.7 cm³

- Stage: T1b

Tab 2: Classification

- Type: Clear Cell RCC

- Confidence: 89%

Tab 3: Growth

- Rate: 0.8 cm/year

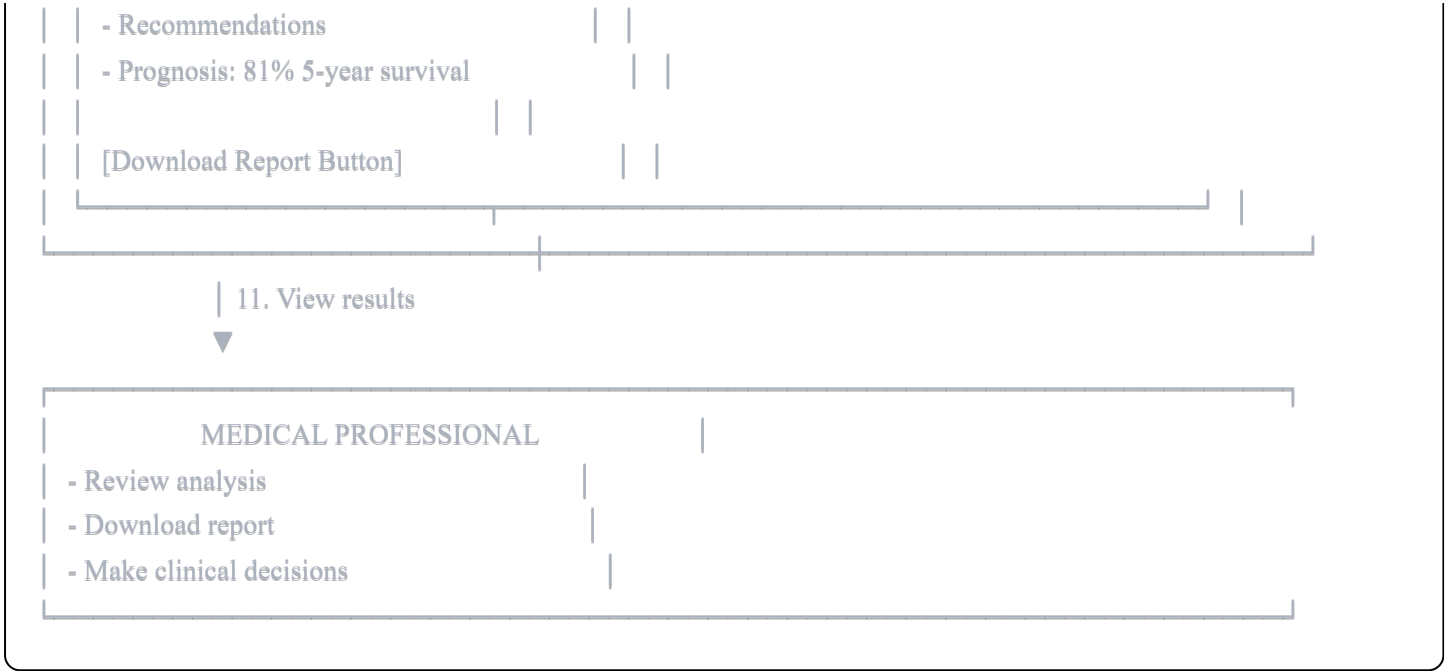
- Doubling: 12-18 months

- Chart showing predictions

Tab 4: Radiomics

- Texture, shape, intensity metrics

Tab 5: Clinical



How to Use These Diagrams

For Documentation:

1. Copy diagrams into your PDF
2. Add to presentation slides
3. Include in README.md

For Presentations:

1. Show overall architecture first
2. Deep dive into training pipeline
3. Demonstrate user flow
4. Explain error handling

For Development:

1. Reference during implementation
2. Debug data flow issues
3. Onboard new team members
4. Plan new features

Converting to Visual Diagrams:

Use these tools to create graphical versions:

- **draw.io** (diagrams.net)
- **Lucidchart**

- **Mermaid** (for markdown)
 - **Microsoft Visio**
 - **PlantUML**
-

Mermaid Code Version (For Markdown)